Numerical Investigation of Brayton Cycle Performance Using Temperature-Dependent Specific Heat Based on NASA Polynomial Thermodynamic Data.

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Abstract

Gas turbine performance is commonly analysed using the Brayton cycle with the assumption of constant specific heat. However, modern gas turbines operate at high turbine inlet temperatures where thermodynamic properties vary significantly with temperature. In this study, a numerical Brayton cycle model was developed in MATLAB to investigate the effect of temperature-dependent specific heat on predicted cycle performance.

Two models were compared: a conventional constant specific heat model and an advanced model incorporating temperature-dependent thermodynamic properties using NASA polynomial correlations. Compressor and turbine exit temperatures were obtained by solving entropy balance equations numerically. The model was used to evaluate net work output, thermal efficiency, back work ratio and optimum pressure ratio across a range of compressor pressure ratios and turbine inlet temperatures representative of modern gas turbines.

The results show that the constant specific heat assumption consistently over-predicts both net work output and thermal efficiency, particularly at high turbine inlet temperatures. Differences of approximately 15–17% were observed in predicted performance parameters. The variable specific heat model provides more realistic thermodynamic behaviour and predicts a shift in optimum pressure ratio. The study demonstrates that temperature-dependent thermodynamic modelling is necessary for accurate preliminary performance analysis of modern gas turbine engines.

Nomenclature

Roman Symbols

BWR Back work ratio
c_p Specific heat at constant pressure (J kg^{-1} K^{-1})
Q_{in} Heat input to the cycle (kJ kg^{-1})
R Specific gas constant (J kg^{-1} K^{-1})
r_p Compressor pressure ratio
T Temperature (K)
T_1 Compressor inlet temperature (K)
T_2 Compressor exit temperature (K)
T_3 Turbine inlet temperature (K)
T_4 Turbine exit temperature (K)
W_c Compressor work (kJ kg^{-1})
W_t Turbine work (kJ kg^{-1})
W_{net} Net work output (kJ kg^{-1})

Greek Symbols

η_{th} Thermal efficiency
γ Ratio of specific heats

Subscripts

c Compressor
t Turbine
net Net cycle value
in Heat input

Constants

a_1, a_2, a_3, a_4, a_5 NASA polynomial coefficients

1. Introduction

Gas turbines are widely used in aircraft propulsion and power generation because of their high power to weight ratio and operational flexibility.Gas turbine performance is commonly analysed using the Brayton cycle model *(Ganesan, 2010)*. The performance of these engines is fundamentally based on the Brayton cycle, which models compression, heat addition, expansion, and heat rejection processes.

Traditional Brayton cycle analysis assumes that air behaves as an ideal gas with constant specific heat. This simplifies calculations and provides acceptable accuracy at low temperatures. However, modern gas turbine engines used in aircraft propulsion often operate at turbine inlet temperatures exceeding 1700K, where thermodynamic property variation becomes significant. Therefore, simplified constant specific heat analysis may lead to inaccurate performance prediction during preliminary engine design.

When temperature effects are ignored, cycle analysis becomes overly optimistic. The compressor work is underestimated and turbine expansion effectiveness is overestimated. As a result, performance predictions such as efficiency and optimum pressure ratio may be inaccurate.

The purpose of this project is to investigate the effect of temperature dependent specific heat on Brayton cycle performance and to quantify the modelling error caused by the constant specific heat assumption.

1.1 Objectives

This study aims to:

- ✓ Develop a MATLAB-based numerical model of the Brayton cycle.
- ✓ Implement temperature dependent specific heat using NASA polynomial relations.
- ✓ Determine compressor and turbine exit temperatures by solving entropy balance equations numerically.

- ✓ Evaluate key performance parameters, including net work output, thermal efficiency, back work ratio, and optimum pressure ratio.
- ✓ Perform a parametric study over a range of compressor pressure ratios and turbine inlet temperatures.
- ✓ Compare the results obtained using constant specific heat and variable specific heat models.
- ✓ Quantify the performance prediction error introduced by the constant specific heat assumption.

1. Literature Review

Gas turbine performance has been studied extensively using the Brayton cycle. In early thermodynamic analysis, the working fluid is usually assumed to behave as an ideal gas with constant specific heat. This assumption simplifies the equations and allows analytical solutions for temperature and efficiency.

However, several researchers have shown that this assumption becomes inaccurate at high turbine inlet temperatures.Advanced gas turbine cycle studies have also shown that constant property assumptions can lead to inaccurate predictions of performance parameters at high temperatures *(Bolland and Stadaas, 1995*). Modern gas turbines operate at temperatures above 1400–1800 K, where the thermodynamic properties of air vary significantly with temperature. In particular, the specific heat increases and the ratio of specific heats decreases as temperature rises.

Previous studies have shown that constant specific heat analysis generally over predicts cycle efficiency and net work output *(Saravanamuttoo et al., 2017)*. The reason is that constant property models underestimate the compressor work and overestimate turbine expansion work. As a result, the predicted optimum pressure ratio and fuel consumption can be unrealistic for modern engines.

Researchers have therefore introduced temperature dependent thermodynamic models using polynomial property relations, such as NASA thermodynamic polynomials. Several studies have also analysed gas turbine cycles using temperature dependent thermodynamic properties to obtain more realistic performance predictions *(Sanaye and Dehghandokht, 2010)*. These models allow the specific heat, enthalpy, and entropy of air to vary with temperature and provide more realistic performance predictions.

Studies also show that the effect of variable specific heat becomes more significant as turbine inlet temperature increases.Similar conclusions have been reported in gas turbine cycle studies that account for temperature dependent thermodynamic properties *(Al-Sulaiman, 2011).* At lower temperatures, constant and variable property models give similar results. However, at high temperatures, the difference becomes large and cannot be neglected.

Although these advanced thermodynamic models exist, many preliminary gas turbine analyses and educational calculations still use constant specific heat assumptions. Therefore, it is useful to quantify how much error is introduced by this simplification and to demonstrate the importance of temperature dependent modelling.

The present study contributes to this area by numerically comparing constant specific heat and variable specific heat Brayton cycle models over a range of pressure ratios and turbine inlet temperatures representative of modern gas turbines.

2. Methodology

Two thermodynamic models were created:

3.1 Baseline Model

Assumes:

- ✓ Ideal gas
- ✓ Constant Specific Heat
- ✓ Analytical Isentropic Relations

This represents the classical Brayton cycle analysis used in textbooks.

3.2 Advanced Model

Uses:

- ✓ Temperature Dependent Specific Heat $c_p(\mathrm{T})$.

- ✓ Entropy Balance Equations.
- ✓ Numerical Integration.
- ✓ Iterative Root Finding Solution.

NASA Polynomial Thermodynamic Data

$$c_p(T) = R(a_1 + a_2T + a_3T^2 + a_4T^3 + a_5T^4)$$

where:

$a_1 \ldots a_5 =$ NASA polynomial coefficiency

These coefficients were obtained from the NASA thermodynamic database *(McBride et al., 2002)* and used to compute the temperature dependent thermodynamic properties of air.

Because c_p varies with temperature, exit temperatures cannot be calculated analytically. Therefore, the compressor and turbine temperatures were solved numerically using entropy equations. MATLAB numerical integration and nonlinear solvers were used to determine the thermodynamic state points and calculate cycle performance.

3. Performance Parameters

The Brayton cycle model was evaluated using several key thermodynamic performance parameters. These parameters allow the effect of pressure ratio and temperature dependent specific heat to be quantified.

1. Net Work Output

The net work output of the cycle is defined as the difference between turbine work and compressor work.

$$W_{net} = W_t - W_c$$

This parameter indicates the useful work produced by the gas turbine.

2. Thermal Efficiency

The thermal efficiency represents the fraction of heat input that is converted into useful work.

$$\eta_{th} = \frac{W_{net}}{Q_{in}}$$

Higher efficiency indicates better energy conversion performance.

3. Back Work Ratio

The back work ratio represents the fraction of turbine work required to drive the compressor.

$$BWR = \frac{W_c}{W_t}$$

Gas turbines typically have relatively high back work ratios compared with other power cycles.

4. Optimum Pressure Ratio

The optimum pressure ratio corresponds to the compressor pressure ratio at which the Brayton cycle produces maximum net work output .For an ideal Brayton cycle operating with constant thermodynamic properties, the compressor pressure ratio that maximizes net work output can be approximated by:

$$r_{p,opt} = \left(\frac{T_3}{T_1}\right)^{\frac{\gamma}{2(\gamma-1)}}$$

This equation shows that the optimum pressure ratio increases with increasing turbine inlet temperature. However, when temperature-dependent specific heat is considered, the optimum pressure ratio shifts to a lower value due to the increase in compressor work and the reduction in turbine expansion effectiveness. This behaviour is observed in the numerical results presented in Figures 1 and 2.

5. Prediction Error

The prediction error was calculated to quantify the difference between results obtained using constant specific heat and temperature dependent specific heat models.

5. Results and Discussion

The numerical results show consistent trends for all turbine inlet temperatures.

1. Net work increases with pressure ratio initially.
2. After a certain point, compressor work dominates.
3. A maximum work output occurs at an optimum pressure ratio.

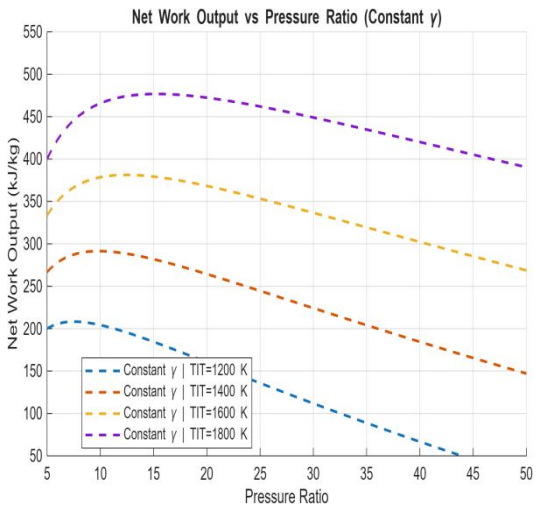

Figure 1: Net work vs pressure ratio (constant c_p)

This figure shows how net work changes with pressure ratio when the specific heat is assumed constant. As the pressure ratio increases, the turbine produces more expansion work, so the net work initially increases. After a certain point, the compressor requires much more work to compress the air. Because of this, the compressor work starts to dominate and the net work decreases. The peak of each curve represents the optimum pressure ratio, where the Brayton cycle produces maximum work.

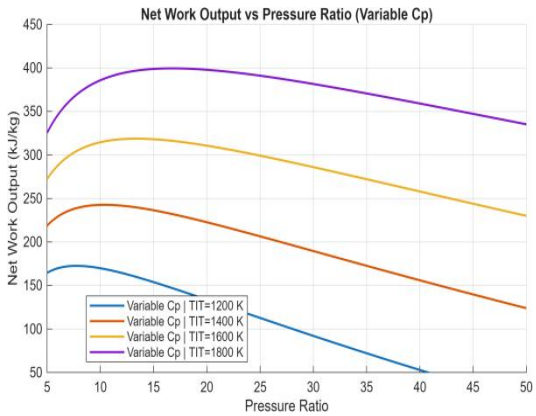

Figure 2: Net work vs pressure ratio (variable c_p)

This figure shows the net work when the specific heat varies with temperature. The general trend is similar: net work increases first and then decreases with pressure ratio. However, the predicted work output is lower than in the constant c_p model. This happens because at high temperatures the specific heat of air increases, which means more energy is required during compression and the turbine expansion becomes less effective. For example, at a pressure ratio of 30 the constant specific heat model predicts a net work output approximately 15–17% higher than the temperature dependent model. As a result, the variable c_p model predicts lower and more realistic work output.

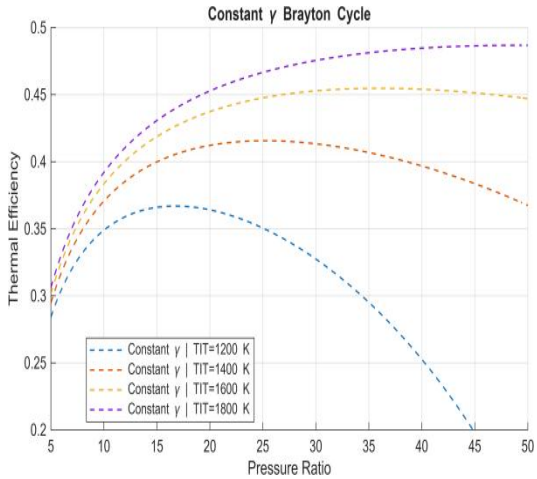

Figure 3: Thermal Efficiency vs Pressure Ratio (Constant c_p)

This graph shows how thermal efficiency changes with pressure ratio when the specific heat is assumed constant. As the pressure ratio increases, the efficiency increases because the Brayton cycle converts heat into useful work more effectively. Higher turbine inlet temperatures also produce higher efficiency because more energy is available for expansion in the turbine. However, at very high pressure ratios the efficiency improvement becomes smaller, especially for lower turbine inlet temperatures.

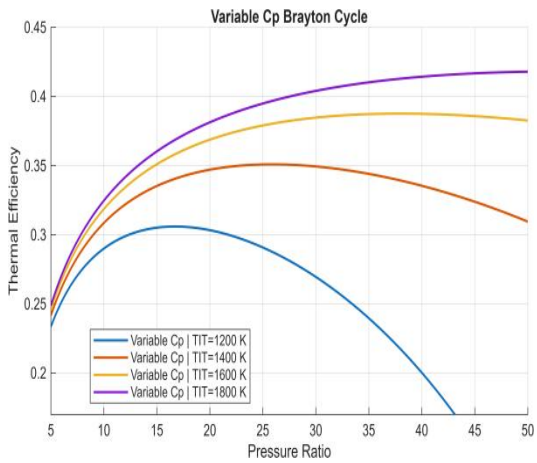

Figure 4: Thermal Efficiency vs Pressure Ratio (Variable c_p)

This graph shows the thermal efficiency when the specific heat varies with temperature. The overall trend is similar, with efficiency increasing as the pressure ratio increases. However, the predicted efficiencies are lower than those from the constant c_p model. This happens because at high temperatures the specific heat of air increases, meaning more energy is required during compression and less useful work is produced during turbine expansion. As a result, the variable c_p model gives more realistic efficiency predictions.

Comparing Figures 3 and 4 shows that the constant specific heat model slightly overestimates the thermal efficiency of the Brayton cycle especially at higher turbine inlet temperatures. For instance, at a pressure ratio of 30 the constant specific heat model overpredicts thermal efficiency by approximately 15%.

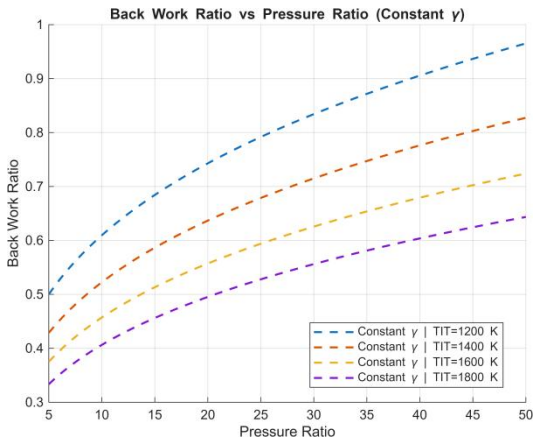

Figure 5: Back work ratio vs pressure ratio (constant c_p)

This graph shows how the back work ratio changes with pressure ratio when the specific heat is assumed constant. As the pressure ratio increases, the compressor requires more work to compress the air to a higher pressure. Because of this, a larger fraction of the turbine work is used just to drive the compressor. This causes the back work ratio to increase with pressure ratio. The curves also show that higher turbine inlet temperatures reduce the back work ratio slightly because the turbine produces more work.

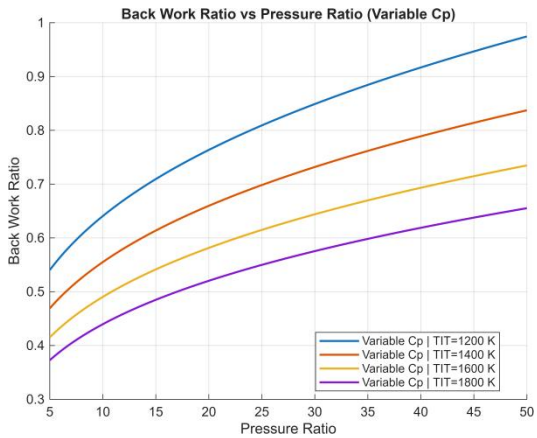

Figure 6: Back work ratio vs pressure ratio (variable c_p)

This graph shows the back work ratio when the specific heat varies with temperature. The trend is similar, with the back work ratio increasing as the pressure ratio increases. However, the values are generally higher than those predicted by the constant c_p model. At higher pressure ratios the difference between the two models becomes smaller, with prediction errors typically below 10%. This occurs because at higher temperatures the specific heat of air increases, which means the compressor requires more energy during compression. As a result, a larger portion of the turbine work is consumed by the compressor.

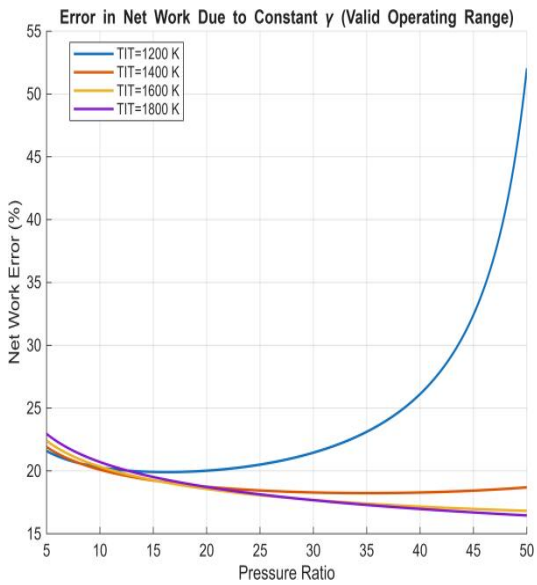

Figure 7: Error in Net Work Due to Constant γ

This graph shows the percentage error in the predicted net work when the Brayton cycle is analysed using constant thermodynamic properties instead of temperature dependent properties. At lower pressure ratios, the error is

smaller because the temperature changes in the cycle are not very large. However, as the pressure ratio increases, the temperature in the compressor and turbine becomes much higher. At these higher temperatures, the properties of air change significantly so the constant property assumption becomes less accurate. As a result, the error in the predicted net work increases, especially for the lower turbine inlet temperature case.

When temperature dependent specific heat is included:

1. Net work output decreases
2. Thermal efficiency decreases
3. Back work ratio increases
4. Optimum pressure ratio shifts

This occurs because at high temperature, the specific heat of air increases. More energy is required to compress the air and turbine expansion becomes less effective. Therefore, the constant specific heat model produces overly optimistic predictions *(Mattingly, 2006)*.

6. Model Validation

To check that the MATLAB Brayton cycle program is working correctly, a validation test was carried out. The results from the variable specific heat model were compared with the constant specific heat model under conditions where both models are expected to behave almost the same.

For this test, a turbine inlet temperature of 1200 K and a compressor pressure ratio of 10 were used. At this relatively low temperature, the specific heat of air does not change very much with temperature. Because of this, the constant specific heat model can be used as a good reference to check the numerical solution.

Validation at Moderate Temperature
Table 1: $T_3 = 1200$ K & r = 10

Parameter	Constant c_p Model	Variable c_p Model	%
Net Work Output	204.2	169.8	16.86%
Thermal Efficiency	0.3489	0.2901	16.86%
Back Work Ratio	0.6094	0.4396	27.86%

Table 1 shows that the constant specific heat model overpredicts the net work output and thermal efficiency of the Brayton cycle. The difference arises because the specific heat of air increases with temperature, which increases compressor work and reduces turbine effectiveness. Consequently, the variable specific heat model provides more realistic predictions of gas turbine performance.

Validation at High Temperature
Table 2: $T_3 = 1800$ K & r = 10

Parameter	Constant c_p Model	Variable c_p Model	%
Net Work Output	465.7	385.8	17.16%
Thermal Efficiency	0.3790	0.3248	14.31%
Back Work Ratio	0.4063	0.4396	8.21%

Table 2 shows the comparison between the constant and temperature dependent specific heat models at a turbine inlet temperature of 1800 K and a pressure ratio of 10. The results indicate that the constant specific heat model overpredicts the net work output and thermal efficiency of the Brayton cycle.

The difference occurs because the thermodynamic properties of air vary significantly at high temperatures. When temperature dependent specific heat is included the compressor requires more work and the turbine expansion becomes less effective.

Table 3: Effect of Turbine Inlet Temperature on Prediction Error (r = 10)

Parameter	Error at 1200K	Error at 1800K
Net Work Output	16.86%	17.16%
Thermal Efficiency	16.86%	14.31%
Back Work Ratio	27.86%	8.21%

Table 3 summarises the prediction error between the constant and variable specific heat models at two different turbine inlet temperatures while maintaining the same pressure ratio. The results

show that the constant specific heat model consistently overpredicts the performance of the Brayton cycle. The difference occurs because the thermodynamic properties of air vary with temperature. At higher turbine inlet temperatures the variation of specific heat becomes more significant, making temperature dependent modelling necessary for realistic gas turbine performance prediction.

Validation at Moderate Temperature
Table 4: $T_3 = 1200$ K & r = 30

Parameter	Constant c_p Model	Variable c_p Model	%
Net Work Output	111.8	92.08	17.67%
Thermal Efficiency	0.3274	0.2696	17.66%
Back Work Ratio	0.8342	0.8489	1.77%

Table 4 presents the comparison between the constant specific heat model and the temperature-dependent specific heat model at a moderate turbine inlet temperature and a pressure ratio of 30. The results show that the constant specific heat model predicts higher net work output and higher thermal efficiency compared with the variable specific heat model.

The net work output predicted by the constant property model is approximately 17.67% higher, while the thermal efficiency is overestimated by about 17.66%. This occurs because the constant specific heat assumption does not account for the increase in specific heat at higher temperatures.

As a result, the compressor work is underestimated and the turbine expansion work is overestimated. The back work ratio shows only a small difference of about 1.77%, indicating that both models predict a similar proportion of turbine work used to drive the compressor at this condition.

Validation at Moderate Temperature
Table 5: $T_3 = 1800$ K & r = 30

Parameter	Constant c_p Model	Variable c_p Model	%
Net Work Output	449.00	381.57	15.02%
Thermal Efficiency	0.4758	0.4041	15.06%
Back Work Ratio	0.5561	0.5960	7.17%

Table 5 shows the comparison between the constant specific heat model and the variable specific heat model at a higher turbine inlet temperature and a pressure ratio of 30. The results show that the constant property model predicts higher net work output and thermal efficiency than the variable property model.

The net work output is overestimated by about 15.02%, and the thermal efficiency by about 15.06%. This happens because the constant specific heat assumption does not account for the increase in specific heat at high temperatures. The back work ratio is slightly higher in the variable specific heat model, showing that more turbine work is required to drive the compressor when temperature effects are included.

Table 6: Effect of Turbine Inlet Temperature on Prediction Error (TIT = 1200)

Parameter	Error at r = 10	Error at r = 30
Net Work Output	16.86%	17.67%
Thermal Efficiency	16.86%	17.66%
Back Work Ratio	27.86%	1.77%

This table compares the percentage error between the constant and variable specific heat models at different pressure ratios. The results show that the constant specific heat assumption consistently overpredicts the net work output and thermal efficiency of the Brayton cycle. The difference occurs because the thermodynamic properties of air vary with temperature. The back work ratio shows a different behaviour because both compressor and turbine work increase with pressure ratio, which reduces the relative difference between the two models.

Validation with Classical Brayton Cycle Theory

To further verify the accuracy of the developed MATLAB Brayton cycle model, the predicted cycle efficiency was compared with the classical thermodynamic relation for an ideal Brayton cycle operating with constant specific heat.

For an ideal Brayton cycle, the thermal efficiency can be expressed as:

$$\eta_{th} = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}}$$

For a pressure ratio of r_p =10 and γ =1.4, the theoretical Brayton cycle efficiency is approximately 0.36.

The MATLAB constant specific heat model predicted an efficiency of 0.3489, which shows good agreement with classical Brayton cycle analysis reported in the literature such as *(Gas Turbine Theory, Saravanamuttoo et al.,2017; Mattingly,2006 and Elements of Propulsion: Gas Turbines and Rockets.)*

7. Physical Interpretation of the Results

The change in performance occurs because the thermodynamic behaviour of air changes at high temperature. At low temperatures, air molecules mainly store energy in translational motion. However, as the temperature increases, additional molecular energy modes become active, especially vibrational energy modes.

Because of this, the internal energy of the air increases faster with temperature. This causes the specific heat to increase. When the specific heat increases, more heat energy is required to produce the same temperature rise during compression therefore, the compressor requires more work.

At the same time, the ratio of specific heats γ decreases as temperature increases. A lower γ reduces the effectiveness of turbine expansion. This means the turbine produces less useful work from the same temperature drop. Since the compressor consumes more work and the turbine produces relatively less work, the net work output decreases. Because thermal efficiency depends on net work output, the overall cycle efficiency also decreases.

The constant specific heat model cannot capture these effects because it assumes fixed thermodynamic properties. As a result, it over-predicts turbine work and under-predicts compressor work, leading to optimistic efficiency predictions.

8. Validation Conclusion

The difference between the two models is therefore not a mistake or numerical error. It is a real physical effect caused by the change of air properties with temperature.
The results also show stable behaviour of the numerical solution and proper convergence of the equations.

This confirms that:

- ✓ the entropy equations were solved correctly.
- ✓ The numerical integration is working properly.
- ✓ The root finding method is stable

Therefore the developed MATLAB Brayton cycle model can be considered valid and reliable especially for analysing higher turbine inlet temperatures where the constant specific heat assumption is no longer accurate.

9. Engineering Meaning

The results show that constant property thermodynamic analysis is acceptable only for low temperature gas turbines. For modern engines operating above 1500K temperature dependent modelling is required for accurate preliminary design.

If constant c_p analysis is used in engine design:

- ✓ Compressor size may be underestimated.
- ✓ Fuel consumption may be underestimated.
- ✓ Operating pressure ratio may be incorrect.

Therefore variable c_p modelling should be used before performing detailed CFD or turbomachinery design.

10. Conclusion

This project investigated the influence of temperature dependent specific heat on Brayton cycle performance using a MATLAB numerical model. Two approaches were compared: the conventional constant specific heat analysis and a variable specific heat model based on NASA polynomial thermodynamic data.

The results showed that the constant specific heat assumption over predicts both thermal efficiency and net work output, especially at high turbine inlet temperatures. When temperature dependent properties were included, the predicted compressor work increased, turbine expansion effectiveness decreased, and the back work ratio became higher. In addition, the optimum pressure ratio shifted from the value predicted by the constant specific heat model. These findings demonstrate that simplified Brayton cycle analysis can lead to unrealistic performance predictions for modern

high temperature gas turbines.

Therefor temperature dependent thermodynamic modelling is necessary for accurate preliminary gas turbine performance evaluation. The developed numerical model provides a more realistic representation of engine behaviour and can serve as a useful preliminary step before detailed turbomachinery design or computational fluid dynamics analysis.

This project provided a strong foundation in numerical thermodynamic modelling and gas turbine performance analysis. It also encouraged further interest in higher fidelity propulsion modelling methods, such as mean-line turbomachinery analysis and computational fluid dynamics, which are important for modern aircraft engine development.

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